Inputs (3 tokens, $d_k = d_v = 2$)

	Q	K	V
t_1	1 0	1 0	1 0
t_2	0 1	1 1	0 1
t_3	1 1	0 1	1 1

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V$$

causal mask: $M_{ij} = 0$ if $j \le i$, $M_{ij} = -\infty$ if $j > i$

A Without mask

1 $QK^\top$

t_1	1	1	0
t_2	0	1	1
t_3	1	2	1

$\div\sqrt{2}$

2 $QK^\top/\sqrt{d_k}$

t_1	0.7071	0.7071	0.0000
t_2	0.0000	0.7071	0.7071
t_3	0.7071	1.4142	0.7071

softmax

3 weights A

t_1	0.4011	0.4011	0.1978
t_2	0.1978	0.4011	0.4011
t_3	0.2483	0.5035	0.2483

$\times V$

4 output AV

t_1	0.5989	0.5989
t_2	0.5989	0.8022
t_3	0.4965	0.7517

B With causal mask

5 $QK^\top/\sqrt{d_k} + M$

t_1	0.7071	$-\infty$	$-\infty$
t_2	0.0000	0.7071	$-\infty$
t_3	0.7071	1.4142	0.7071

softmax

6 masked weights

t_1	1.0000	0.0000	0.0000
t_2	0.3302	0.6698	0.0000
t_3	0.2483	0.5035	0.2483

$\times V$

7 masked output

t_1	1.0000	0.0000
t_2	0.3302	0.6698
t_3	0.4965	0.7517

What the mask changes: t_1 attends only to itself, so its output is v_1. t_2 loses t_3 and renormalises over t_1, t_2. t_3 is unchanged: as the last token it could already see every earlier token.

Values rounded to 4 decimals and computed with src.attention (unit-tested against these numbers).
Hatched cells are masked: $-\infty$ before softmax, so their weight is exactly 0.